

Homework Assignment #2

Deliverables

In this assignment, students will:

- Create and modify matrices in MATLAB.
- Use matrix indexing to extract rows, columns, and submatrices.
- Use matrix concatenation to organize engineering data.
- Use matrix-generation functions and reshaping tools where appropriate.
- Work with 3D arrays and reshaping.
- Provide clear output using appropriate variable names and comments.

Questions 1–3 should be submitted to **MATLAB Grader** and will be evaluated mainly on the **functionality and correctness of the code**. These questions are equally weighted and amount to a total of **10 points** toward the assignment grade.

It is recommended that you write and debug your code in MATLAB on your own computer before submitting it to MATLAB Grader. You will have **50 submission attempts** on MATLAB Grader.

Question 4 should be submitted to Crowdmark as a PDF and to the LEARN Dropbox as a MATLAB script file. It will be evaluated for both code functionality and readability.

To submit Question 4 to Crowdmark, please generate a PDF document that includes:

1. A header comment block at the beginning of your code, formatted as follows:

```
% Author: Your Name
% Purpose: Short description of what the code does
% Date Created: Date
% Acknowledgements: Any outside sources used to complete
% assignment, including websites, books, and people
% other than the teaching team
```

2. The MATLAB code that you wrote.
3. The command window output from your code, including all input and output statements.
4. Any requested written explanations.

Code files (.m files) for all four questions should also be submitted to the **LEARN Dropbox** to help the teaching team check code structure, readability, and plagiarism.

Please name your files using the following format:

```
hwX_qX_FirstInitialLastName.m
```

For example:

hw2_q1_JTaheriShakib.m
hw2_q2_JTaheriShakib.m
hw2_q3_JTaheriShakib.m
hw2_q4_JTaheriShakib.m

Your LEARN Dropbox submission should contain the same code that you submitted to MATLAB Grader and Crowdmark.

Suggested General Instructions for Students

Please follow these instructions for all questions:

1. Use clear and meaningful variable names.
2. Include comments in your code to explain important steps.
3. Do not manually calculate results that MATLAB can calculate for you.
4. Use matrix indexing whenever the question asks you to extract or modify part of a matrix.
5. Check that your matrix dimensions are correct before submitting.
6. For MATLAB Grader questions, make sure the required variable names are exactly the same as those given in the question.
7. For Question 4, use `fprintf` to display your final results clearly.
8. For Question 4, include a short written explanation in complete sentences.
9. Submit your `.m` files to the LEARN Dropbox.
10. Make sure your Crowdmark PDF includes the code, command window output, and written explanation.

Question 1

Problem Description

A field engineer collected water quality data from four sampling locations along a river. The measured parameters are:

1. Nitrate concentration, mg/L
2. Phosphate concentration, mg/L
3. Turbidity, NTU

The data are provided below:

```
nitrate = [14.2 9.8 16.5 11.4];  
phosphate = [0.32 0.18 0.41 0.25];  
turbidity = [8.5 5.2 12.6 7.4];
```

Each value in the vectors represents one sampling location.

Required MATLAB Tasks

Write a MATLAB script that creates the following variables:

- a) `waterMat`: a 4×3 matrix where each row represents one sampling location and the columns are nitrate, phosphate, and turbidity.
- b) `station2`: a row vector containing all measurements for Sampling Location 2.
- c) `phosphateCol`: a column vector containing only the phosphate measurements.
- d) `highRiskSubMat`: a 2×2 matrix containing nitrate and turbidity data for Sampling Locations 3 and 4.
- e) `waterMat_updated`: a copy of `waterMat` where the turbidity value at Sampling Location 3 is replaced by 10.0.
- f) `waterMat_noPhosphate`: a matrix created by deleting the phosphate column from `waterMat`.
- g) `lastValue`: the last value in `waterMat` using the `end` operator.
- h) `linearVals`: a vector containing elements 3 to 6 of `waterMat` using linear indexing.

What you need to submit

Copy your completed code to MatLab Grader and press the 'Submit' button to check your work. If the code is correct, no further action is needed. If there are still some bugs, you have the opportunity to resubmit your code for this problem up to 50 times. Your final script file should be submitted to the LEARN Dropbox.

Question 2

Problem Description

A geotechnical investigation was conducted at five boreholes. For each borehole, the following soil properties were measured:

1. Depth to bedrock, m
2. Standard Penetration Test value, SPT-N
3. Moisture content, %
4. Unit weight, kN/m³

Use the following starter script:

```
depth = [6.5 8.2 5.9 10.1 7.4];  
spt = [18 25 14 32 21];  
moisture = [12.5 15.2 18.1 10.8 14.6];  
unitWeight = [18.4 19.1 17.8 20.2 18.9];
```

Each vector contains data for Boreholes 1 to 5.

Required MATLAB Tasks

Write a MATLAB script that creates the following variables:

- a) `geoMat`: a 5×4 matrix where each row represents one borehole.
- b) `borehole3`: a row vector containing all data for Borehole 3.
- c) `sptValues`: a column vector containing only the SPT-N values.
- d) `firstThreeBoreholes`: a 3×4 matrix containing all properties for Boreholes 1 to 3.
- e) `geoMat_modified`: a copy of `geoMat` where the SPT-N value for Borehole 2 is replaced by 28.
- f) `geoMat_noMoisture`: a matrix created by deleting the moisture-content column.
- g) `oddBoreholes`: a matrix containing Boreholes 1, 3, and 5 only.
- h) `geoMat_extra`: a new matrix where an additional column of zeros is added to the right side of `geoMat`. This column will later be used for a calculated stability index.

What you need to submit

Copy your completed code to MATLAB Grader and press the 'Submit' button to check your work. If the code is correct, no further action is needed. If there are still some bugs, you have the opportunity to resubmit your code for this problem up to 50 times.

Question 3

Problem Description

A monitoring program measured contaminant concentrations at four locations over two sampling days. Three contaminants were measured:

1. Chloride, mg/L
2. Sulfate, mg/L
3. Nitrate, mg/L
4. Iron, mg/L

The data for Day 1 and Day 2 are:

```
day1 = [120 85 14 0.30;  
        135 92 18 0.42;  
        110 78 11 0.25];  
  
day2 = [125 88 13 0.28;  
        142 96 20 0.45;  
        118 82 12 0.27];
```

Rows represent sampling locations. Columns represent contaminants.

Required MATLAB Tasks

Write a MATLAB script that creates the following variables:

- a) `contam3D`: a $3 \times 4 \times 2$ array. Page 1 should contain `day1`, and Page 2 should contain `day2`.
- b) `day1_loc2`: a row vector containing all contaminant values for Location 2 on Day 1.
- c) `day2_nitrate`: a column vector containing nitrate values for all locations on Day 2.
- d) `iron_loc3_day2`: the iron concentration at Location 3 on Day 2.
- e) `subArray`: a 2×2 matrix containing chloride and sulfate data for Locations 1 and 2 on Day 1.
- f) `contam3D_updated`: a copy of `contam3D` where the nitrate value at Location 2 on Day 2 is replaced by 19.
- g) `allValuesVector`: a column vector containing all values from `contam3D` using linear indexing.
- h) `reshapedData`: a 4×6 matrix created by reshaping `contam3D`.

What you need to submit

Copy your completed code to MATLAB Grader and press the 'Submit' button to check your work. If the code is correct, no further action is needed. If there are still some bugs, you have the opportunity to resubmit your code for this problem up to 50 times.

Question 4

Problem Description

A landfill cover system is monitored to evaluate ground settlement over time. Four settlement plates are installed across the site. The vertical settlement, in mm, is measured at three different times:

1. Month 1
2. Month 3
3. Month 6

The measured settlement data are shown below:

Settlement Plate	Month 1	Month 3	Month 6
Plate 1	4.2	8.5	14.1
Plate 2	3.8	7.9	13.5
Plate 3	5.1	10.2	18.4
Plate 4	2.9	6.8	11.7

Use the following starter script:

```
settlement = [4.2 8.5 14.1;  
              3.8 7.9 13.5;  
              5.1 10.2 18.4;  
              2.9 6.8 11.7];  
  
plateID = [1; 2; 3; 4];
```

Required MATLAB Tasks

Write a MATLAB program that:

- Finds the size of the `settlement` matrix.
- Extracts the Month 6 settlement values into a column vector called `month6`.
- Extracts all settlement data for Plate 3 into a row vector called `plate3Data`.
- Calculates the total settlement increase from Month 1 to Month 6 for each plate and stores it in `increase`.
- Creates a new matrix called `settlementReport` by concatenating `plateID`, `settlement`, and `increase`.

The final matrix should have the following columns:

```
Plate ID | Month 1 | Month 3 | Month 6 | Increase
```

- Finds the plate with the largest settlement increase.
- Replaces the Month 6 settlement for Plate 4 with 12.0 in a new matrix called `settlement_corrected`.

- h) Deletes the Month 1 column from `settlement_corrected` and stores the result in `recentSettlement`.
- i) Uses `fprintf` to display a clear summary similar to:

```
The largest settlement increase occurred at Plate X.  
The maximum settlement increase was Y mm.
```

Written Explanation

In two or three sentences, explain why matrix indexing is useful for organizing environmental and geological monitoring data.

What you need to submit

The code, command window output, and written explanation should be submitted in a single PDF file to Crowdmark. Your script file should be submitted to the LEARN Dropbox.